

NYCU Memory and Storage System, Homework 1

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Q0. (10%) Example

Please run `q0.fio` on your own device and write down your output `q0` in the report. Also specify the type of drive that is used(HDD/SSD...).

Ans:

Device Type: External SSD connected via USB 3.1 Gen2

Output:

```
Run status group 0 (all jobs):
  READ: bw=27.4MiB/s (28.7MB/s), 27.4MiB/s-27.4MiB/s (28.7MB/s-28.7MB/s), io=1024MiB (1074MB), run=37394-37394msec
```

Explanation:

Although the device is an SSD connected through a USB 3.1 port, the measured bandwidth is only ~ 28.9 MiB/s. In my opinion, this is because the test uses `bs=4k`, `iodepth=1`, and `direct=1`, which represents a **synchronous small-block sequential read** pattern.

In this case, throughput is bottlenecked by per-request latency ($\sim 135 \mu\text{s}$) rather than interface bandwidth. The theoretical maximum speed can be estimated as: $4\text{KB} \div 135\mu\text{s} \approx 29 \text{ MB/s}$, which closely matches the measured result.

Q1. (10%) Read vs. Randread

Is there any significant difference between read/randread? Why or why not? Please justify your answer in brief.

Ans:

There is no significant difference between sequential read and random read in this case.

Output:

```
Run status group 0 (all jobs):
  READ: bw=26.6MiB/s (27.9MB/s), 26.6MiB/s-26.6MiB/s (27.9MB/s-27.9MB/s), io=64.0MiB (67.1MB), run=2407-2407msec

Run status group 1 (all jobs):
  READ: bw=29.0MiB/s (30.4MB/s), 29.0MiB/s-29.0MiB/s (30.4MB/s-30.4MB/s), io=64.0MiB (67.1MB), run=2208-2208msec
```

Explanation:

Unlike HDDs, which rely on spinning disk and moving read head, SSDs have no moving parts and can access any location at nearly the same speed regardless of access pattern. Therefore, both sequential and random reads yield similar bandwidth (~27–29 MiB/s) and average latency (~134–146 μ s).

Summary:

Item	Sequential Read	Random Read
Bandwidth	~27.9 MB/s	~30.4 MB/s
IOPS	6806	7420
Avg. Latency	146.14 μ s	133.73 μ s

Q2. (10%) Write vs. Randwrite

Is there any significant difference between write/randwrite? Why or why not? Please justify your answer in brief.

Ans:

There is no significant difference between sequential write and random write.

Output:

```
Run status group 0 (all jobs):
  WRITE: bw=54.7MiB/s (57.3MB/s), 54.7MiB/s-54.7MiB/s (57.3MB/s-57.3MB/s), io=64.0MiB (67.1MB), run=1171-1171msec

Run status group 1 (all jobs):
  WRITE: bw=51.6MiB/s (54.1MB/s), 51.6MiB/s-51.6MiB/s (54.1MB/s-54.1MB/s), io=64.0MiB (67.1MB), run=1241-1241msec
```

Explanation:

Similar to the read results in Q1, since the device is an SSD, there is no mechanical seek time penalty for random access. Both sequential

and random writes yield similar bandwidth (~52–55 MiB/s) and average latency (~71–75 μ s).

There is another thing is **write speed (~54 MiB/s) is significantly faster than read speed (~27 MiB/s)**. This is because writes are buffered in the SSD's internal cache and acknowledged quickly, whereas reads with `direct=1` must wait for actual data retrieval from the storage medium.

Summary:

Item	Sequential Write	Random Write
Bandwidth	~57.3 MB/s	~54.1 MB/s
IOPS	14,000	13,200
Avg. Latency	70.99 μ s	75.01 μ s

Q3. (10%) Forward write vs. Backward write

Is there any significant difference between forward/backward write? Why or why not? Please justify your answer in brief.

Ans:

There is a little difference between forward write and backward write.

Output:

```
Run status group 0 (all jobs):  
  WRITE: bw=60.8MiB/s (63.8MB/s), 60.8MiB/s-60.8MiB/s (63.8MB/s-63.8MB/s), io=64.0MiB (67.1MB), run=1052-1052msec  
  
Run status group 1 (all jobs):  
  WRITE: bw=53.1MiB/s (55.6MB/s), 53.1MiB/s-53.1MiB/s (55.6MB/s-55.6MB/s), io=64.0MiB (67.1MB), run=1206-1206msec
```

Explanation:

Forward write is faster (~60.8 MiB/s) compared to backward write (~53.1 MiB/s), **with about 13% higher bandwidth and lower average latency** (63.76 μ s vs. 73.10 μ s).

In my opinion, this is because most SSDs are internally optimized for sequential forward access. The SSD's firmware and write buffer are designed to prefetch and pipeline forward writes more efficiently.

Summary:

Item	Forward Write	Backward Write
Bandwidth	~63.8 MB/s	~55.6 MB/s
IOPS	15,600	13,600
Avg. Latency	63.76 μ s	73.10 μ s

Q4. (15%) Buffered read vs. Nonbuffered read

1. (10%) Is there any significant difference between buffered/nonbuffered sequential read? Why or why not? Please justify your answer in brief.

Ans:

There is a huge difference between buffered and non-buffered reads.

Output:

```
Run status group 0 (all jobs):
  READ: bw=941MiB/s (987MB/s), 941MiB/s-941MiB/s (987MB/s-987MB/s), io=64.0MiB (67.1MB), run=68-68msec

Run status group 1 (all jobs):
  READ: bw=27.8MiB/s (29.2MB/s), 27.8MiB/s-27.8MiB/s (29.2MB/s-29.2MB/s), io=64.0MiB (67.1MB), run=2302-2302msec
```

Explanation:

Since stonewall ensures buffered read runs first, the 64MB test file is loaded into the OS page cache (RAM) during the first job. When fio reads the data, it is served directly from RAM, resulting in extremely high bandwidth and very low latency; with direct=1, the OS page cache is bypassed completely. Every read request goes directly to the SSD through the USB interface.

Summary:

Item	Buffered Read	Non-buffered Read
Bandwidth	~1017 MB/s	~26.4 MB/s
IOPS	248,000	6,452
Avg. Latency	3.78 μ s	154.24 μ s

2. (5%) Replace sequential read with random read. Is there any significant difference between buffered/nonbuffered random read? Why or why not? Please justify your answer in brief.

Ans:

There is no significant difference between buffered and non-buffered random reads in this case.

Output:

```
Run status group 0 (all jobs):
  READ: bw=27.3MiB/s (28.6MB/s), 27.3MiB/s-27.3MiB/s (28.6MB/s-28.6MB/s), io=64.0MiB (67.1MB), run=2343-2343msec

Run status group 1 (all jobs):
  READ: bw=27.4MiB/s (28.7MB/s), 27.4MiB/s-27.4MiB/s (28.7MB/s-28.7MB/s), io=64.0MiB (67.1MB), run=2339-2339msec
```

Explanation:

In my opinion, cache is not warmed up for random access. In the sequential buffered read test, fio reads the file from start to end, which naturally loads the entire file into the OS page cache before the second job runs. However, with random reads, the access pattern is unpredictable, and the page cache is not fully populated before the non-buffered job starts. Therefore, the buffered randread does not benefit from cached data as effectively.

Summary:

Item	Buffered Randread	Non-buffered Randread
Bandwidth	~28.6 MB/s	~28.7 MB/s
IOPS	6,992	7,004
Avg. Latency	142.09 μ s	141.20 μ s

Q5. (10%) LBA

Is there any bandwidth trend between these jobs? Why or why not?
Please justify your answer in brief. You can also experiment with the size of the file.

Ans:

There is no significant bandwidth trend across different offsets in these jobs.

Output:

```
Run status group 0 (all jobs):  
  READ: bw=736MiB/s (771MB/s), 736MiB/s-736MiB/s (771MB/s-771MB/s), io=1024MiB (1074MB), run=1392-1392msec  
  
Run status group 1 (all jobs):  
  READ: bw=752MiB/s (789MB/s), 752MiB/s-752MiB/s (789MB/s-789MB/s), io=1024MiB (1074MB), run=1361-1361msec  
  
Run status group 2 (all jobs):  
  READ: bw=750MiB/s (787MB/s), 750MiB/s-750MiB/s (787MB/s-787MB/s), io=1024MiB (1074MB), run=1365-1365msec  
  
Run status group 3 (all jobs):  
  READ: bw=765MiB/s (802MB/s), 765MiB/s-765MiB/s (802MB/s-802MB/s), io=1024MiB (1074MB), run=1339-1339msec  
  
Run status group 4 (all jobs):  
  READ: bw=743MiB/s (779MB/s), 743MiB/s-743MiB/s (779MB/s-779MB/s), io=1024MiB (1074MB), run=1379-1379msec
```

Explanation:

Because the device used is an SSD, which has no mechanical components. Unlike HDDs, where data stored on outer tracks (lower offset) can be read faster due to higher linear velocity, and data on inner tracks (higher offset) is read slower, SSDs access any location at the same speed regardless of the logical block address.

Job	Offset	Bandwidth
job 1	80%	771 MB/s
job 2	60%	789 MB/s
job 3	40%	787 MB/s
job 4	20%	802 MB/s
job 5	0%	779 MB/s

Summary:

Q6. (15%) Blocksize

1. (10%) Is there any significant difference between 4k/1k read? Why or why not? Please justify your answer in brief.

Ans:

Yes, there is a significant difference between 4k and 1k block size read.

Output:

```
Run status group 0 (all jobs):  
  READ: bw=26.7MiB/s (28.0MB/s), 26.7MiB/s-26.7MiB/s (28.0MB/s-28.0MB/s), io=64.0MiB (67.1MB), run=2394-2394ms  
  
Run status group 1 (all jobs):  
  READ: bw=7457KiB/s (7636kB/s), 7457KiB/s-7457KiB/s (7636kB/s-7636kB/s), io=64.0MiB (67.1MB), run=8789-8789ms
```

Explanation:

With bs=1k, each I/O request only transfers 1KB of data, but still have the same per-request latency overhead ($\sim 133 \mu\text{s}$) as bs=4k. To read the same 64MB of data, bs=1k requires 4x more I/O requests.

Since latency per request is roughly the same, total time is $\sim 4x$ longer, resulting in $\sim 4x$ lower bandwidth

Item	bs=4k	bs=1k
Bandwidth	26.7 MiB/s (28.0 MB/s)	7.27 MiB/s (7.64 MB/s)
IOPS	6,843	7,456
Average Latency	145.21 μs	133.32 μs
Total I/O requests	16,384	65,536
Total time	2,394 ms	8,789 ms

Summary:

2. (5%) If you want to achieve the best performance in the condition above, how would you modify blocksize? Explain it briefly.

Ans:

To achieve the best performance, increase the block size as large as possible.

Since per-request latency remains roughly constant regardless of block size, using a larger block size means more data is transferred per request, which directly increases bandwidth.

Q7. (20%) Fastest nonbuffered read

Please explain how you achieve the fastest 1G nonbuffered read in brief.

Ans:

To achieve the fastest 1G nonbuffered read, the key parameter is block size. By testing block sizes from bs=1k to bs=2m, a clear and consistent trend is observed: larger block size leads to higher bandwidth.

Output:

```
Run status group 0 (all jobs):
  READ: bw=871MiB/s (914MB/s), 871MiB/s-871MiB/s (914MB/s-914MB/s), io=1024MiB (1074MB), run=1175-1175msec

Run status group 1 (all jobs):
  READ: bw=748MiB/s (784MB/s), 748MiB/s-748MiB/s (784MB/s-784MB/s), io=1024MiB (1074MB), run=1369-1369msec

Run status group 2 (all jobs):
  READ: bw=678MiB/s (711MB/s), 678MiB/s-678MiB/s (711MB/s-711MB/s), io=1024MiB (1074MB), run=1510-1510msec

Run status group 3 (all jobs):
  READ: bw=621MiB/s (652MB/s), 621MiB/s-621MiB/s (652MB/s-652MB/s), io=1024MiB (1074MB), run=1648-1648msec

Run status group 4 (all jobs):
  READ: bw=495MiB/s (519MB/s), 495MiB/s-495MiB/s (519MB/s-519MB/s), io=1024MiB (1074MB), run=2068-2068msec

Run status group 5 (all jobs):
  READ: bw=343MiB/s (359MB/s), 343MiB/s-343MiB/s (359MB/s-359MB/s), io=1024MiB (1074MB), run=2988-2988msec

Run status group 6 (all jobs):
  READ: bw=143MiB/s (150MB/s), 143MiB/s-143MiB/s (150MB/s-150MB/s), io=1024MiB (1074MB), run=7149-7149msec

Run status group 7 (all jobs):
  READ: bw=89.8MiB/s (94.2MB/s), 89.8MiB/s-89.8MiB/s (94.2MB/s-94.2MB/s), io=1024MiB (1074MB), run=11401-11401msec

Run status group 8 (all jobs):
  READ: bw=47.2MiB/s (49.5MB/s), 47.2MiB/s-47.2MiB/s (49.5MB/s-49.5MB/s), io=1024MiB (1074MB), run=21690-21690msec

Run status group 9 (all jobs):
  READ: bw=27.5MiB/s (28.8MB/s), 27.5MiB/s-27.5MiB/s (28.8MB/s-28.8MB/s), io=1024MiB (1074MB), run=37249-37249msec

Run status group 10 (all jobs):
  READ: bw=7476KiB/s (7655kB/s), 7476KiB/s-7476KiB/s (7655kB/s-7655kB/s), io=1024MiB (1074MB), run=140264-140264msec
```

Bonus (10%)

1. (5%) Compare with multiple kinds of storage devices

Is there any significant difference between read/randread on storage devices?
Why or why not? Please justify your answer in brief.

Ans: None

2. (5%) Find the specs of your own hardware that you tested on

Is your hardware running to spec? If not, could you come up with a possible theory?

Ans:

My hardware is “Kingston XS1000 External SSD”

Spec	Official Rating
Interface	USB 3.2 Gen 2 (10 Gbps)
Sequential Read	Up to 1,050 MB/s
Sequential Write	Up to 1,000 MB/s

Not fully — the device reaches about 87% of its rated speed at best, and significantly lower for small block sizes.

Actual Test Results vs. Official Specs

Test	Official Spec	Actual Result	% of Spec
Q7 best read (bs=2m)	1,050 MB/s	≈ 914 MB/s	87%
Q5 read (bs=1m)	1,050 MB/s	≈ 812 MB/s	77%
Q0 read (bs=4k)	1,050 MB/s	≈ 30 MB/s	3%

In summary, the device performs close to spec (~87%) under optimal conditions (bs=2m), but real-world performance depends heavily on block size, queue depth, and I/O pattern.